Name: ____________________

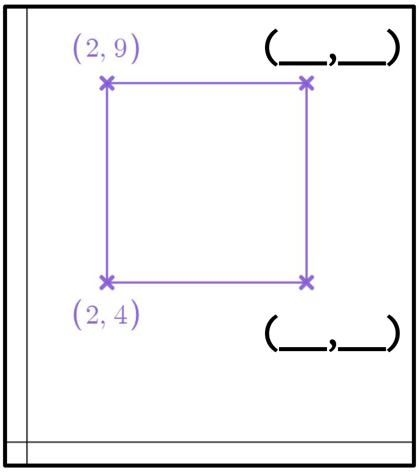
Worksheet

Problem solving with plotting coordinates

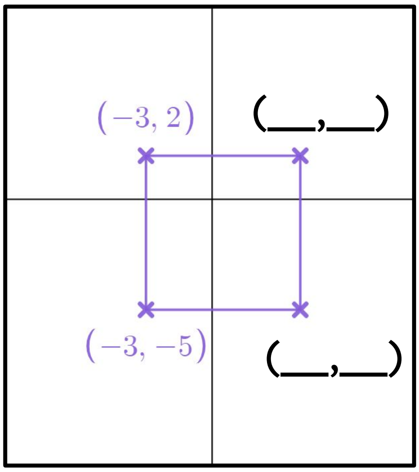
Task A: Calculating horizontal and vertical distance

1) Work out the missing coordinates to make each shape a square.

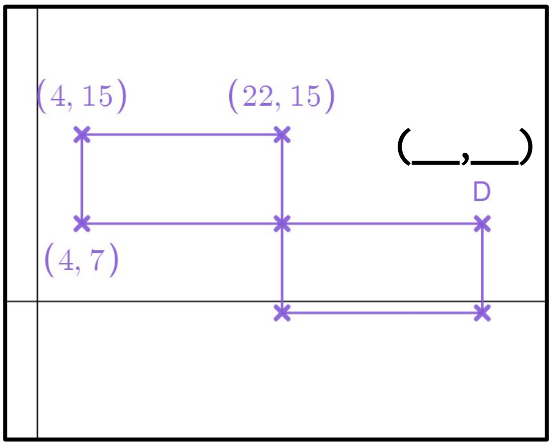
a)



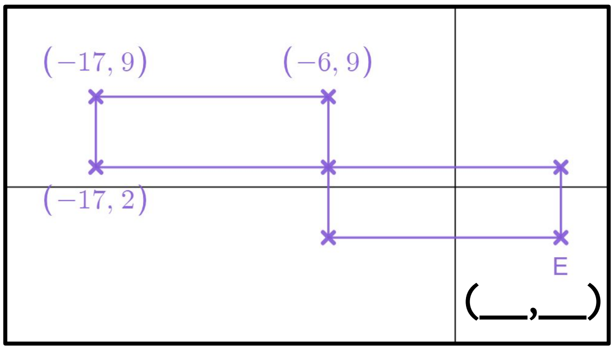
b)



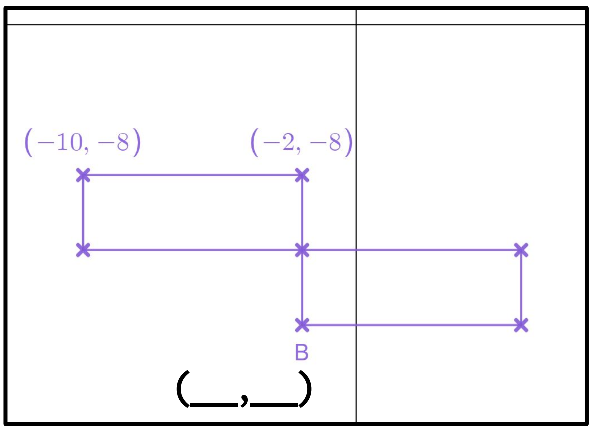
2a) These two rectangles are the same size. Work out the missing coordinate D.



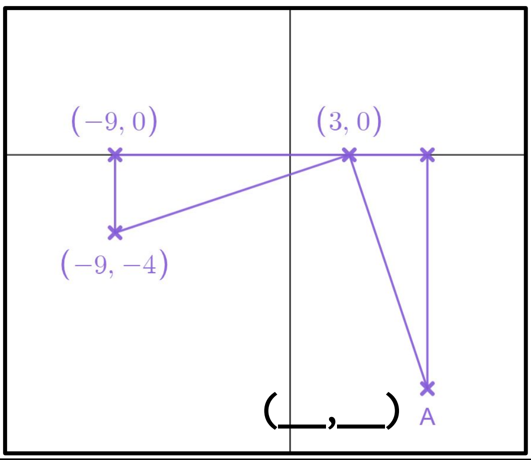
2b) These two rectangles are the same size. Work out the missing coordinate E.



2c) These two rectangles are the same size. The length is double the width. Work out the missing coordinate B.



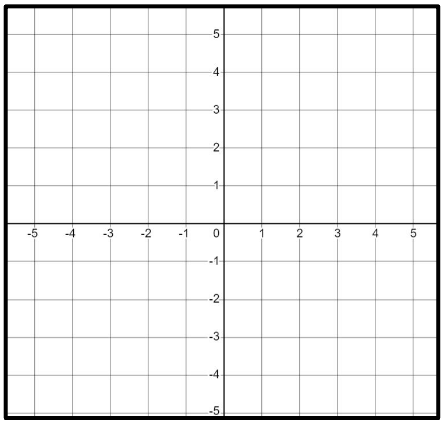
2d) Below are two identical triangles. Work out the missing coordinate A.

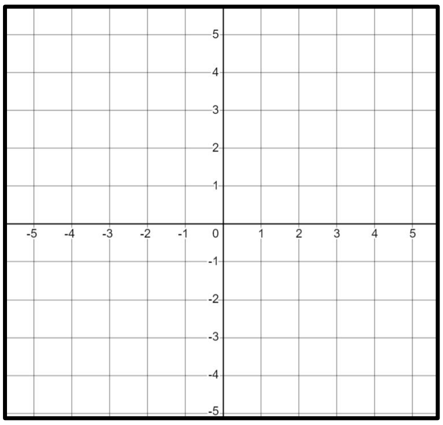


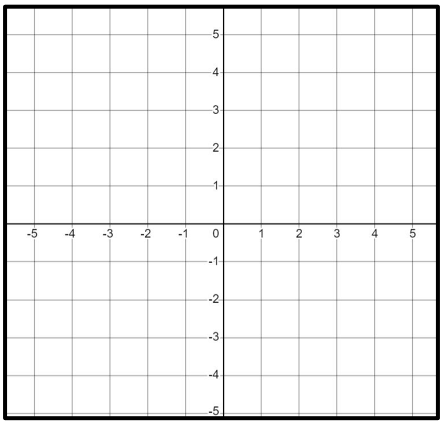
Task B: Finding the coordinates where two lines intersect

1) Find a coordinate that follows both rules. You may wish to use the axes to help.

a) $x = -3\ ,\ y = 4$ b) $x = 2\ ,\ y = -1$ c) $x = -3\ ,\ y = 0$

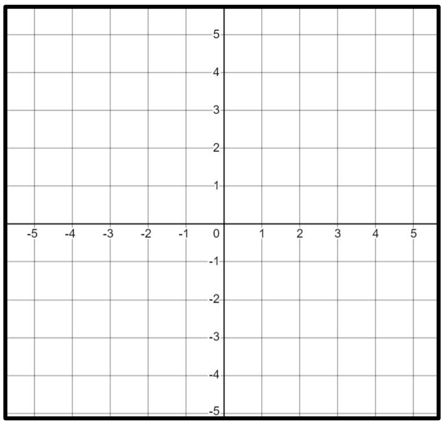


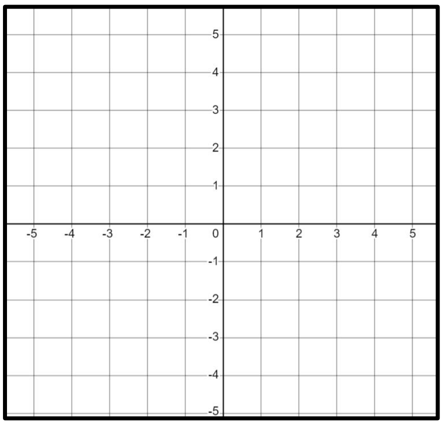


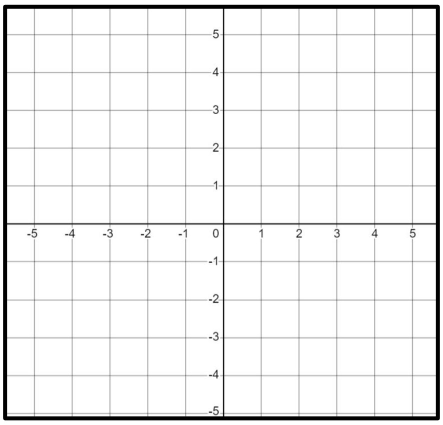


1) Find a coordinate that follows both rules. You may wish to use the axes to help.

d) $x = 2,\ y = 2x$ e) $y = x,\ \ y = 4$ f) $y = 3x,\ y = x + 2$







2) Where do each pair of lines intersect?
(list some coordinates that fit the rules to help you)

a) $y = x + 5, \quad x = 4$

b) $y = 2x, \quad y = x + 1$

c) $y = 2x, \quad y = x + 3$

d) $y = 2x, \quad y = x + 5$

e) $y = 2x, \quad y = x - 1$

f) $y = 2x, \quad y = x$

g) $x + y = 6, \quad y = x$

h) $y = -x, \quad y = x - 4$

i) $y = -x, \quad y = x + 4$

j) $y = 4x, \quad y = x + 6$

k) $y = \frac{1}{2}x, \quad y = x + 3$